$$\alpha = \frac{1.5\times10^{-3}\ \text{S m}^2\ \text{mol}^{-1}}{300\times10^{-4}\ \text{S m}^2\ \text{mol}^{-1}}$$

$$\alpha = 0.05$$

$$K_a = \frac{\alpha^2 c}{1-\alpha}$$

$$= \frac{(0.05)^2(0.01)}{1-0.05}$$

$$= \frac{25\times10^{-4}\times10^{-2}}{95\times10^{-2}}$$

$$= 0.26\times10^{-4}$$

$$= 2.6\times10^{-5}.$$

13. Given

$I = 1.608A$; $t = 50\ \text{min} = 50\times60 = 3000S$

$\eta = 100\%$

$V = 250\ \text{mL}$

$C = 0.5\ \text{M}$

Calculate the number of faradays of electricity passed through the $CuSO_4$ solution

$\Rightarrow Q = It$

$Q = 1.608\times3000$

$Q = 4824C$

$\therefore$ number of Faradays of electicity $= \frac{4824\ \text{C}}{96500\ \text{C}} = 0.05\text{F}$

Electrolysis of $CuSO_4$

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s).$

The above equation shows that 2F electricity will deposit 1 mole of Cu^{2+} to Cu.

$\therefore$ 0.05F electricity will

deposit $\frac{1\text{mol}}{2\text{F}}\times0.05\text{F} = 0.025\ \text{mol}$

Initial number of molar of Cu^{2+} in 250 ml of solution $= \frac{0.5}{1000\ \text{mL}}\times250\text{mL}$

$= 0.125\ \text{mol}$

$\therefore$ number of moles of Cu^{2+} after electrolysis $= 0.125 - 0.025$

$= 0.1\ \text{mol}$

$\therefore$ Concentration of $Cu^{2+} = \frac{0.1\ \text{mol}}{250\ \text{mL}}\times1000\ \text{mL}$

$= 0.4\ \text{M}$

14. Required half cell reaction

$2\,Br^- \rightarrow Br_2 + 2e^-$ $\quad (E^\circ_{ox}) = -1.09V$

$2\,Fe^{3+} + 2e^- \rightarrow 2Fe^{2+}$ $\quad (E^\circ_{red}) = +0.771V$

$2Fe^{3+} + 2Br^- \rightarrow 2Fe^{2+} + Br_2$ $\quad (E^\circ_{cell}) = ?$

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